



Smart bin (Credit: Hackster.io)

need of the hour for monitoring the environment and tackling climate change. Miniaturised environmental sensors like gas, chemical and biosensors help in remote sensing of the environment. Integrated environmental-economic information is vital for policymakers. The increasing role of remote sensing devices for on-road screening and environment sensing helps in enforcing emission control policies.

Areas for monitoring environmental safety

Solar energy monitoring. It detects the amount of solar radiation to determine the actual performance of the solar power plant. Real-time analytics are required for scheduling operation and maintenance tasks like cleaning and orientation, demand and supply measurement, net generated and unused electricity measurement, real-time performance ratio, solar yield calculation and active monitoring for faults that are difficult to detect.

The aim of solar energy monitoring is to send immediate notifications in case of any malfunction by monitoring each device of the plant. It is based on the Internet of Things (IoT) and Big Data technologies that offer scalability for centralised monitoring of millions of devices with a lot of computations running at the same time at a single point,

regardless of location and technology. Energy meters and other devices send data directly to the cloud via IoT data loggers. These also divert excess electricity generated to specific appliances, enabling intelligent use of solar power.

Weather monitoring. Weather information like temperature, pressure, humidity, soil moisture, wind speed and direction, precipitation amount, UV index, snow depth and solar radiation is gathered simultaneously from a weather monitoring system. Observations are taken automatically whenever required.

Data loggers enable weather sensors to collect data and transmit through mobile or another monitoring device over GPRS/GSM network, making real-time information available at fingertips, irrespective of distance. The observatory can issue a warning signal to the public immediately after poor weather conditions are recognised. Low-power sensors, user configuration through SMSes, over-the-air (OTA) firmware upgrades and secure operations make a weather monitoring system versatile and user-friendly.

Climate monitoring. It integrates satellite observations, ground-based data and forecast models to monitor and forecast changes in the weather and climate. Measurements are taken over time for statistical analysis and

identification of mean values, trends and variations. The more the available information, the better the climate can be understood, and the more accurately future conditions can be assessed. It allows monitoring climate-controlled environments in places like museums, labs, etc. It measures and warns of risks arising from rainfall, landslides, floods, storms and so on.

Water quality monitoring. As demand for water grows and population pressure increases, quality of surface water becomes an increasing concern worldwide. Water quality is also affected by environmental change. Researchers and engineers worldwide are trying to develop models capable of predicting the effects of such changes. Various high-resolution, multi-parameter water quality measuring systems have been developed using innovative miniaturised chemical sensing devices. These systems aim to convert contaminated water into clean water that is fit for use. These also monitor the water quality of rivers, lakes, streams and other water bodies using appropriate control systems and probes.

Road monitoring. Ice formation, collected water and other conditions that may affect road safety are monitored by specific stations. Traffic data can be measured using sensors, detectors and other techniques according to their applications, performance and costs. Detectors that help achieve this are pneumatic instruments, inductive loops, magnetometers, piezoelectric sensors, microwave detectors, infrared detectors, ultrasound detectors, passive acoustic array sensors and video detection systems.

Air monitoring. It measures gas concentrations in closed or open areas and the quality of air for breathing using specific measuring systems. An ambient air-quality monitoring system monitors the level of pollutants to improve air quality index (AQI). Monitoring air quality by major public and private bodies is essential to preserve health and assess emission sources.

Waste monitoring. It monitors landfills, proper waste disposal and recycling using sensors. Waste management technology helps track the